

$$\int \frac{2x_A}{N_A - x_A(N_A + N_B)}$$

$$\int \frac{\ln(N_A - x_A(N_A + N_B))}{N_A + N_B}$$

$$\frac{\ln(N_A - x_A N_A - x_A N_B)}{N_A + N_B}$$

$$\frac{\ln N_A - \ln x_A N_A - \ln x_A N_B}{N_A + N_B}$$

$$\cancel{\ln N_A} - (\cancel{\ln x_A} + \cancel{\ln N_A}) - (\ln x_A + \ln N_B)$$

$$- \ln x_A - \ln x_A + \ln N_B$$

$$\ln N_B - 2 \ln x_A$$

18.4.1

Prove the following equation using definitions in Table 18.4-1

$$N_A + N_B = C V_m$$

$$\Rightarrow N_A = J_A^* + C_A V_m$$

$$J_A^* + C_A V_m + J_B^* + C_B V_m = C V_m$$

$$\Rightarrow J_A^* + J_B^* = 0$$

$$(C_A V_m + C_B V_m = C V_m$$

$$\Rightarrow (C_A + C_B) V_m = C V_m$$

$$\Rightarrow C_A + C_B = C \text{ (overall concentration)}$$

$$\Rightarrow \boxed{C V_m = C V_m}$$

18.4.5

Gas A is diffusing from a gas stream at point 1 to a catalyst surface at point 2 & reacts instantaneously & irreversibly as follows: $2A \rightarrow B$

Gas B diffuses back to the gas stream

Derive the final equation for N_A at constant P & S.S. in terms of partial pressures

$$N_A = J_A^* + C_A V_m \quad J_A^* = -C D_{AB} \frac{\partial x_A}{\partial z}$$

$$N_A = -C D_{AB} \frac{\partial x_A}{\partial z} + C_A V_m$$

$$C_A = x_A C$$

$$N = C V_m = N_A + N_B$$

$$V_m = \frac{N_A + N_B}{C}$$

$$N_A = -C D_{AB} \frac{\partial x_A}{\partial z} + C_A \left(\frac{N_A + N_B}{C} \right)$$

$$N_A = -C D_{AB} \frac{\partial x_A}{\partial z} + \frac{C_A}{C} (N_A + N_B)$$

$$N_A = -C D_{AB} \frac{\partial x_A}{\partial z} + x_A (N_A + N_B)$$

$$\frac{N_A - x_A (N_A + N_B)}{x_A} = -C D_{AB} \frac{\partial x_A}{\partial z}$$

$$\int \frac{\partial x_A}{N_A - x_A (N_A + N_B)} = \int \frac{\partial z}{-C D_{AB}}$$

$$\int \frac{\partial x_A}{N_A - x_A N_A - x_A N_B} = \frac{(z_2 - z_1)}{-C D_{AB}}$$

it's not complete

18.1.1

CH₄ & He gas contained in a tube @ 101.32 kPa & 298 K. At one point, partial pressure of methane is $P_{A1} = 10.79$ kPa & at a point 0.02 m distance away, $P_{A2} = 20.26$ kPa. If the total pressure is constant throughout the tube, calculate the flux of CH₄ at S.S. for equimolar counterdiffusion

$$J_{A2}^* = -D_{AB} \frac{dC_A}{dz}$$

$$D_{AB} = 0.675 \text{ (m}^2/\text{s)} \cdot 10^{-4}$$

→ steady state means J_{A2}^* & D_{AB} constant

$$J_{A2}^* \int_{z_1}^{z_2} dz = -D_{AB} \int_{C_1}^{C_2} dC_A$$

$$J_{A2}^* (z_2 - z_1) = -D_{AB} (C_{A2} - C_{A1})$$

$$J_{A2}^* = \frac{D_{AB} (C_{A1} - C_{A2})}{z_2 - z_1}$$

→ since total pressure constant, C is constant, & can use $PV = nRT$

$$PV = nRT \rightarrow \frac{n}{V} = \frac{P}{RT} = C$$

$$\rightarrow C_{A1} = \frac{P_{A1}}{RT} \quad C_{A2} = \frac{P_{A2}}{RT}$$

$$\rightarrow J_{CH_4}^* = \frac{D_{AB} (P_{A1} - P_{A2})}{RT (z_2 - z_1)}$$

$$J_{CH_4}^* = \frac{0.675 \times 10^{-4} \text{ m}^2/\text{s}}{5} \frac{(10.79 \times 10^3 - 20.26 \times 10^3 \text{ Pa}) \cdot \text{kg mole}}{(8314.3 \text{ m}^3 \cdot \text{Pa}) (298 \text{ K}) (0.02 \text{ m})}$$

$$J_{CH_4}^* = 5.52 \times 10^{-5} \frac{\text{kg mole CH}_4}{\text{s} \cdot \text{m}^2}$$

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8.2.1

FOR A mixture of ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) vapor & methane (CH_4), predict the diffusivity using the method of Fuller et al.

A) At $1.0132 \times 10^5 \text{ Pa}$ $\rightarrow 1 \text{ atm}$ & 298 & 373 K

$$D_{AB} = \frac{1.00 \times 10^{-7} T^{1.75} (1/M_A + 1/M_B)^{0.5}}{P [\sum V_A^{1/3} + \sum V_B^{1/3}]^2}$$

$$M_A = (2 \cdot 12.01) + (6 \cdot 1.008) + 16 = 46.068$$

$$M_B = 12.01 + (4 \cdot 1.008) = 16.042$$

$$\sum V_A = (2 \cdot 16.5) + (6 \cdot 1.98) + 5.48 = 50.36$$

$$\sum V_B = (16.5) + (4 \cdot 1.98) = 24.42$$

$$D_{AB} = \frac{(1.00 \times 10^{-7})(298)^{1.75} (1/46.068 + 1/16.042)^{0.5}}{(1) [(50.36)^{1/3} + (24.42)^{1/3}]^2}$$

$$D_{AB}(298) = 1.425 \times 10^{-5} \text{ m}^2/\text{s}$$

$$D_{AB} = \frac{(1 \times 10^{-7})(373)^{1.75} (1/46.068 + 1/16.042)^{0.5}}{(1) [(50.36)^{1/3} + (24.42)^{1/3}]^2}$$

$$D_{AB}(373) = 2.111 \times 10^{-5} \text{ m}^2/\text{s}$$

B) At $2.0265 \times 10^5 \text{ Pa}$ & 298 K
 $\rightarrow 2 \text{ atm}$

$$D_{AB} = \frac{(1.00 \times 10^{-7})(298)^{1.75} (1/46.068 + 1/16.042)^{0.5}}{(2) [(50.36)^{1/3} + (24.42)^{1/3}]^2}$$

$$= 6.1962694 \times 10^{-6}$$

$$D_{AB}(2 \text{ atm}, 298 \text{ K}) = 7.125 \times 10^{-6} \text{ m}^2/\text{s}$$

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2.5

Dilute NaOH is diffusing in an aqueous solution
Do as follows:

A.) estimate the diffusivity of NaOH @ 25°C
Also, calculate the diffusivity of the
individual ions Na⁺ & OH⁻

$$D_{AB}^0 = 8.928 \times 10^{-10} T \left(\frac{1/n_+ + 1/n_-}{1/z_+ + 1/z_-} \right)$$

$$D_{AB}^0 = 8.928 \times 10^{-10} (298.15 \text{ K}) \left(\frac{1/1 + 1/2}{1/50.1 + 1/197.6} \right)$$

$$D_{AB}^0 = 1.5958 \times 10^{-5} \text{ cm}^2/\text{s}$$

$$D_{Na^+} = 2.662 \times 10^{-7} \frac{z_i}{n_i}$$

$$= 2.662 \times 10^{-7} \left(\frac{50.1}{1} \right) \Rightarrow D_{Na^+} = 1.334 \times 10^{-5} \frac{\text{cm}^2}{\text{s}}$$

$$D_{OH^-} = 2.662 \times 10^{-7} \left(\frac{197.6}{2} \right) \Rightarrow D_{OH^-} = 2.63 \times 10^{-5} \frac{\text{cm}^2}{\text{s}}$$

$$D_{AB}^0 = \frac{n_1 + n_2}{n_2/D_1 + n_1/D_2} = \frac{1 + 2}{\frac{2}{1.334 \times 10^{-5}} + \frac{1}{2.63 \times 10^{-5}}}$$

$$D_{AB}^0 = 1.596 \times 10^{-5} \text{ cm}^2/\text{s}$$

B.) estimate Diffusivity of NaOH @ 15°C

$$\frac{T}{344 \mu_w}$$

where μ_w = viscosity of water in
cp at new T app. A.2

$$\frac{(15 + 273.15)}{(344)(1.41005)} (1.5958 \times 10^{-5})$$

$$D_{AB}(15^\circ\text{C}) = 1.348 \times 10^{-5} \text{ cm}^2/\text{s}$$

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